



Solving equations

AN EQUATION IS A MATHEMATICAL STATEMENT THAT CONTAINS AN EQUALS SIGN.

Equations can be rearranged to find the value of an unknown variable, such as x or y .

Simple equations

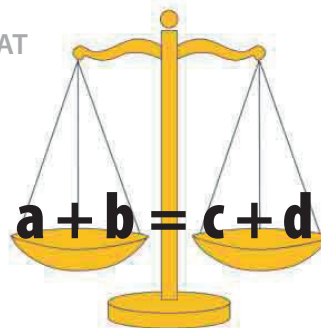
Equations can be rearranged to find the value of an unknown number, or variable. A variable is represented by a letter, such as x or y . Whatever action is taken on one side of an equation must also be made on the other side, so that both sides remain equal.

To find the value of x the equation must be rearranged so that x is by itself on one side of the equation.

Changes made to one side of the equation must also be made to the other side. Subtract 2 from both sides to isolate x .

Simplify the equation by canceling out the $+2$ and -2 on the left side. This leaves x on its own on the left.

Once x is the subject of the equation, working out the right side of the equation gives the value of x .



Left-hand side

Right-hand side

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◀ 168–169 What is algebra?

◀ 172–173 Working with expressions

◀ 177–179 Formulas

Linear graphs 182–185 ▶

◀ Balancing

The expressions on either side of the equals sign in an equation are always equal.

to get rid of this 2, 2 must also be taken from the other side

$$2 + x = 8$$

this expression has the same value as the expression on the other side of the equals sign

subtract 2 on this side

$$2 + x - 2 = 8 - 2$$

2 was subtracted from the other side, so it must also be subtracted from this side

cancel out $+2$ and -2 , which gives 0

$$x = 8 - 2$$

x is now the subject of the equation

$$x = 6$$

working out the right side of the equation ($8 - 2$) gives the value of (6)

LOOKING CLOSER

Creating an equation

Equations can be created to explain day-to-day situations. For example, a taxi firm charges \$3 to pick up a customer, and \$2 per mile traveled. This can be written as an equation.

pick-up cost

$$c = 3 + 2d$$

total cost of the trip

cost per mile multiplied by distance

total cost of trip

$$18 = 3 + 2d$$

cost per mile multiplied by distance

pick-up cost

$$15 = 2d$$

3 has been taken from this side

3 has been taken from this side

this side has been divided by 2

$$7\frac{1}{2} \text{ mi} = d$$

to get rid of 2 in $2d$, divide both sides by 2

Substitute the cost of the trip into the equation.

Rearrange the equation – subtract 3 from both sides.

Find the distance traveled by dividing both sides by 2.

If a customer pays \$18 for a trip, the equation can be used to work out how far the customer traveled.